2022

AUSTRALIAN MATHEMATICS COMPETITION

AMC.

AUSTRALIAN MATHS TRUST

Upper Primary Grades 5–6

TIME ALLOWED

60 minutes

Instructions and Information 考试说明

General 一般说明

1. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 000 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect answer.
   试卷包含 25 道单项选择题和 5 道填空题，填空题答案均为 000-999 之间的整数。题目设置由易到难，答错不扣分。
2. You may use any teaching aids normally available in your classroom, such as MAB blocks, counters, currency, calculators, play money etc. You are allowed to work on scrap paper and teachers may explain the meaning of words in the paper. Mobile phones, computers or tablets are not permitted.
   考生允许携带简单无编程计算器和常用教具，例如：算数积木、筹码、货币、玩具钞票、便条纸等，监考老师可以为考生解释试卷上出现的单词含义，考生不允许携带手机、电脑或者平板。
3. Diagrams are NOT drawn to scale. They are intended only as aids.
   图形只作为辅助参考，并未按照实际比例绘制。
4. This is not a test so do not worry if you can't answer all the questions. However, try to answer as many as you can — you do not lose marks for incorrect answers.
   这不是一场普通的测验，所以即使不能回答所有问题也不必担心。不过要尽可能回答更多的题目，答错也不会丢掉分数的！
5. Check the details on your answer sheet carefully.
   务必仔细检查答题卡上填写的所有信息。
6. When your teacher gives the signal, begin working on the problems.
   监考老师发出作答指令后，方可开始答题。

Integrity of the competition

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

Reminder

You may sit this competition once, in one division only, or risk no score.

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The answer sheet 答题卡说明

Your answer sheet will be scanned. To make sure the scanner reads your paper correctly, there are some DOs and DON'Ts:

你的答题卡将通过扫卡机进行扫描，为确保扫卡机准确扫描你的答案，请务必遵守以下 " 六要 " 和 " 二不要 " ：

DO 要：

- Mark your exam code and answers with a **2B** pencil.
  使用 2B 铅笔正确填涂准考证号和答案。
- Record your answers on the answer sheet (not on the question paper).
  将答案全部涂在答题卡上（在试卷上作答无效）。
- For questions 1–25, fully colour the bubble matching your answer.
  将题目 1–25 的答案按照填涂示例正确涂在对应方格中。
- For questions 26–30, write your answer in the boxes and fully colour the bubble mathching your answer.
  将题目 26–30 的答案写在方框中，并按照填涂示例正确涂在对应方格中。
- Use an eraser if you want to change an answer or remove any marks or smudges.
  请保持答题卡清洁，修改时用橡皮擦干净。
- Check the number of the answer you are filling in is the same as the number of the question you are answering.
  检查答题卡上填涂的答案数量和作答的题目数量是否一致。

DO NOT 不要：

- Doodle or write anything extra on the answer sheet.
  不要在答题卡上乱写乱画。
- Go outside the red boundaries of the answer sheet.
  不要超出答题卡红色虚线框。

Upper Primary Division

1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

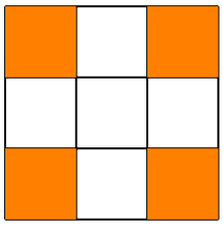
1. 请问哪一个数是二十万五千一百五十？

What number is two hundred and five thousand, one hundred and fifty?

(A) 150 (B) 205 (C) 20 150 (D) 25 150 (E) 205 150

2. 请问右图中阴影部分占总面积的几分之几？

What fraction of this picture is shaded?



(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{9}$ (E) $\frac{5}{9}$

3. 算式 $2220 - 2022 =$

$2220 - 2022 =$

(A) 18 (B) 188 (C) 198 (D) 200 (E) 202

4. 小欧将以下三个数按照从小到大的顺序写下来：

1.03 0.08 0.4

请问小欧写下数字的顺序是什么？

Audrey wrote these three numbers in order from smallest to largest:

1.03 0.08 0.4

In which order did she write them?

(A) $0.08, 1.03, 0.4$ (B) $0.08, 0.4, 1.03$ (C) $0.4, 0.08, 1.03$
(D) $0.4, 1.03, 0.08$ (E) $1.03, 0.4, 0.08$

5. 弟弟满 3 岁时，我 7 岁。请问他满 7 岁时，我的年龄是多少岁？

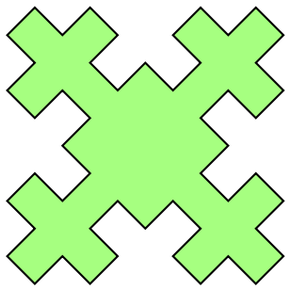
I was 7 years old when my brother turned 3. How old will I be when he turns 7?

(A) 9 (B) 10 (C) 11 (D) 12 (E) 13

6. 右侧图案是由 29 个 1 cm × 1 cm 的小正方形组成的。请问整个图案的周长是多少 cm？

This shape is built from 29 squares, each 1 cm × 1 cm. What is its perimeter in centimetres?

(A) 52 (B) 58 (C) 60 (D) 68 (E) 72



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7. 汽车转速表反映发动机曲轴的转动速度，以每分钟千转计算 (rpm)。如右图所示，请问这个转速表的读数是多少？

A tachometer indicates how fast the crankshaft in a car's engine is spinning, in thousands of revolutions per minute (rpm). What is the reading on the tachometer shown?

(A) 2.2 rpm (B) 2.4 rpm (C) 240 rpm (D) 2200 rpm (E) 2400 rpm

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8. 小周有一个装满 1 升水的瓶子。他喝掉其中的 320 毫升水，请问还剩下多少水？

Joseph had a full, one litre bottle of water. He drank 320 millilitres of it. How much was left?

(A) 660 mL (B) 670 mL (C) 680 mL (D) 730 mL (E) 780 mL

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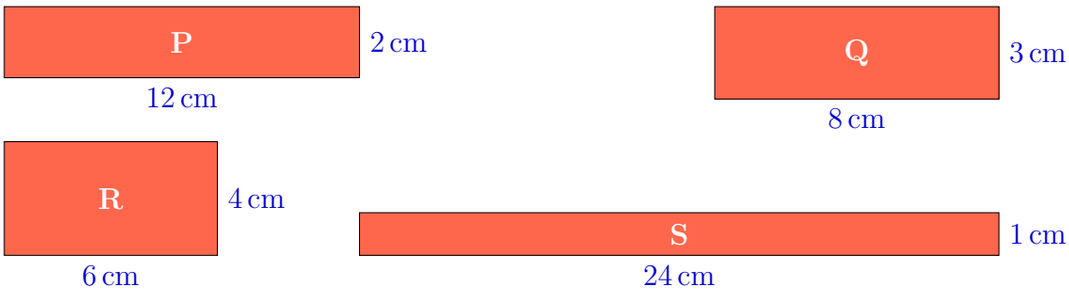
9. 请问下方哪些矩形的面积为 24 平方厘米？



(A) 只有 Q (B) 只有 Q 与 R (C) 只有 R (D) 只有 S (E) P、Q、R 与 S

Which of these rectangles have an area of 24 square centimetres?

(A) Q only (B) Q and R only (C) R only (D) S only (E) P, Q, R and S

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10. 这是小嘉早上的活动安排表。
他必须在 8:55 am 前到达学校，请问他最晚要在什么时间开始洗澡？

(A) 7:35 am (B) 7:50 am (C) 8:05 am (D) 8:20 am (E) 8:35 am

活动	用时
洗澡	15 分钟
穿衣服	10 分钟
吃早餐	20 分钟
刷牙	5 分钟
走路到学校	30 分钟

This table shows Jai's morning routine.
If he needs to be at school by 8:55 am what is the latest time he can start his shower?

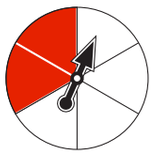
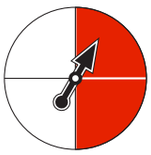
(A) 7:35 am (B) 7:50 am (C) 8:05 am (D) 8:20 am (E) 8:35 am

Activity	Time taken
Shower	15 minutes
Get dressed	10 minutes
Have breakfast	20 minutes
Brush teeth	5 minutes
Walk to school	30 minutes

11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 请问哪一个选项中的指针落到红色的可能性是落到白色的两倍？

Which spinner is twice as likely to land on red as white?

(A)
(B) 
(C) 
(D)
(E)

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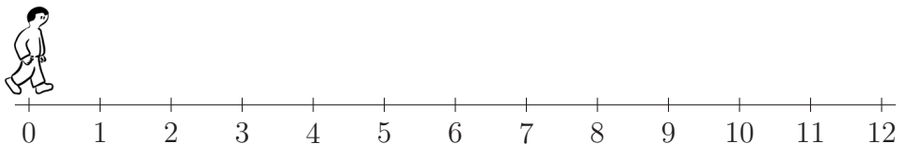
12. 从数轴上的 0 开始，小克按照下列方式来来回回地走：先向数轴的右侧走 3 个单位长度，再向左走 2 个单位长度，接着向右走 3 个单位，再向左走 2 个单位，以此类推。



请问他一共经过表示 $4\frac{1}{2}$ 的位置多少次？

Starting at 0 on the number line, Clement walks back and forth in the following pattern: 3 to the right, 2 to the left, 3 to the right, 2 to the left, and so on.
How many times does he walk past the position represented by $4\frac{1}{2}$?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

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13. 右侧的加法竖式中缺失了三个数字。
小托算出了这三个未知数字并将它们相加。
请问他得到的和是多少？

$$\begin{array}{cccc} & & 4 & \square \\ & 3 & \square & 4 \\ + & \square & 9 & 1 \\ \hline 1 & 0 & 0 & 1 \end{array}$$

Three digits are missing from this sum.
Toby worked out the missing numbers and added them together.
What was his answer?

(A) 11 (B) 13 (C) 15 (D) 17 (E) 19

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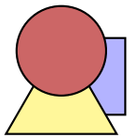
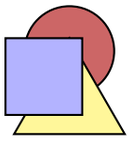
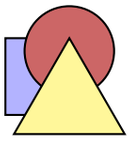
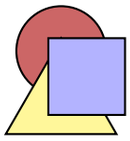
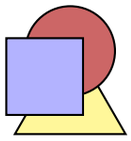
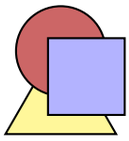
14. 我有三张卡纸，形状分别为正方形、圆形和三角形。
如右图所示，我将这三张卡纸上下叠放并粘贴，
然后将这个粘贴在一起的形状翻面。请问翻面后看到的形状是什么？



I have three cardboard shapes: a square, a circle and a triangle. I glue them on top of each other as shown in this diagram.
I then flip the glued-together shapes over. What could they look like?

(A) 
(B) 
(C) 
(D) 
(E) 

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15. 请问方框内应填入哪个数字才能使下面的等式成立？

$$270 \div 45 = \square \div 15$$

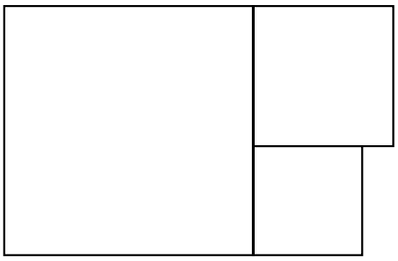
What is the missing number needed to make this number sentence true?

$$270 \div 45 = \square \div 15$$

(A) 3 (B) 6 (C) 60 (D) 90 (E) 150

---

16. 三个大小不同的正方形按照右图所示的方式进行排列。已知最大正方形的周长为 32 cm，最小正方形的面积为 9 cm^2。
请问中等大小的正方形的周长是多少？



Three different squares are arranged as shown. The perimeter of the largest square is 32 centimetres. The area of the smallest square is 9 square centimetres.
What is the perimeter of the medium-sized square?

(A) 12 cm (B) 14 cm (C) 20 cm (D) 24 cm (E) 30 cm

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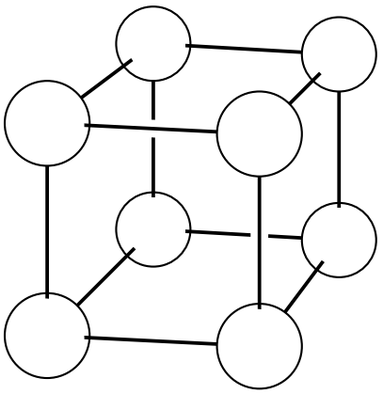
17. 小黄有一袋珠子，小梅取出其中的三分之一。接着小黄再取出剩下珠子的二分之一，最后袋子里面还剩下8 颗珠子。
请问袋子里原来一共有多少颗珠子？

Huang has a bag of marbles. Mei takes out one-third of them. Huang then takes out one-half of those left, leaving 8 marbles in the bag.
How many marbles were originally in the bag?

(A) 12 (B) 16 (C) 18 (D) 24 (E) 36

---

18. 在立方体的各个顶点放置一个互不相同的正整数。
且立方体上同一条边相连的两个数之差都不等于 1。
请问这八个数之和的最小可能值是多少？



A different positive whole number is placed at each vertex of a cube.
No two numbers joined by an edge of the cube can have a difference of 1.
What is the smallest possible sum of the eight numbers?

(A) 36 (B) 37 (C) 38 (D) 39 (E) 40

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19. 乔治今年 78 岁。他有三个孙子：麦克、汤姆和露西。麦克今年 27 岁，汤姆 23 岁，露西 16 岁。请问多少年后乔治的年龄会与他三个孙子年龄的总和相等？

George is 78 this year. He has three grandchildren: Michaela, Tom and Lucy. Michaela is 27, Tom is 23 and Lucy is 16. After how many years will George's age be equal to the sum of his grandchildren's ages?

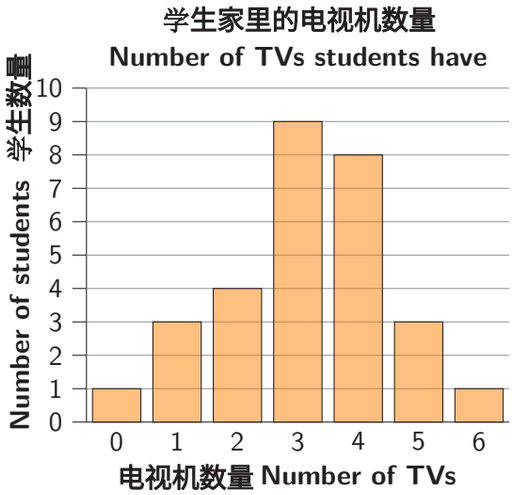
(A) 3 (B) 6 (C) 9 (D) 10 (E) 12

---

20. 葛老师调查自己 5 年级班上的学生每人家中有多少台电视。
右图显示调查的结果。
请问所有学生家中一共有多少台电视？

Ms Graham asked each student in her Year 5 class how many television sets they each have.
This graph shows the results.
How many television sets do the students have altogether?

(A) 9 (B) 29 (C) 91 (D) 99 (E) 101



21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 在一次数学比赛中，共有 70 个男孩和 80 个女孩参加。
最终有 6 个男孩和 15% 的女孩获奖。
请问获奖学生的百分比是多少？

In a mathematics competition, 70 boys and 80 girls competed.
Prizes were won by 6 boys and 15% of the girls.
What percentage of the students were prize winners?

(A) 10% (B) 12% (C) 15% (D) 18% (E) 20%

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22. 小丽将英文字母写在一张纸上，如右图所示。
她将这张纸上下颠倒，然后在浴室的镜子里观察这些字母。请问小丽可以观察到多少个字母的外形完全没有改变？

ABCDEFGHI
JKLMNOPQ
RSTUVWXYZ

Ariel writes the letters of the alphabet on a piece of paper as shown.
She turns the page upside down and looks at it in her bathroom mirror.
How many of the letters appear unchanged when viewed this way?

(A) 0 (B) 3 (C) 4 (D) 6 (E) 9

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23. 澳大利亚数学大学共有 1000 名学生。每名学生每天都要上 6 堂课，每位教师每天都要教 5 堂课且每堂课有 25 名学生。
请问澳大利亚数学大学一共有多少位教师？

The Australian Mathematical College (AMC) has 1000 students. Each student takes 6 classes a day. Each teacher teaches 5 classes per day with 25 students in each class. How many teachers are there at the AMC?

(A) 40 (B) 48 (C) 50 (D) 200 (E) 240

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24. 下方字母组合

$$pqrs, \quad pqsr, \quad prqs, \quad prsq, \ \ldots$$

可以继续排列下去，包含四个字母 p、q、r、s 的排列方式一共会有 24 种。将这些排列方式按照英文字母表的顺序列出。
请问第 19 种字母排列方式是下列哪一种？

This list

$$pqrs, \quad pqsr, \quad prqs, \quad prsq, \ \ldots$$

can be continued to include all 24 possible arrangements of the four letters p, q, r and s. The arrangements are listed in alphabetical order.
Which one of the following is 19th in this list?

(A) $spqr$ (B) $srpq$ (C) $qpsr$ (D) $qrps$ (E) $rpsq$

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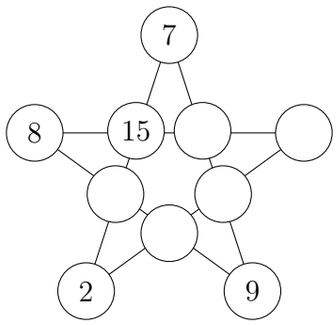
25. 右图是一个益智游戏，每个圆圈内都包含一个整数。图中共有五条直线，每条直线上的四个圆圈内的数字之和都必须等于 40。完成这个游戏后，请问用到的最大数字是多少？



In this puzzle, each circle should contain an integer. Each of the five lines of four circles should add to 40. When the puzzle is completed, what is the largest number used?

(A) 15 (B) 16 (C) 17 (D) 18 (E) 19

问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答案卡上对应的位置。
第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。

For questions 26 to 30, the answer is an integer from 000 to 999.
Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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26. 小阮根据以下规则写下了一些数。
从 1 开始，他将这个数翻倍再加 4，因此他写下的第二个数为 6。
接着他重复这个过程，从最后写下的数开始将它翻倍再加 4，但是如果得到的数大于 100，则不写出这个数的百位数。
请问小阮写下的第 2022 个数是多少？

Nguyen writes down some numbers according to the following rules.
Starting with the number 1, he doubles the number and adds 4, so the second number he writes is 6.
He now repeats this process, starting with the last number written, doubling and then adding 4, but he doesn't write the hundreds digit if the number is bigger than 100.
What is the 2022nd number that Nguyen writes down?

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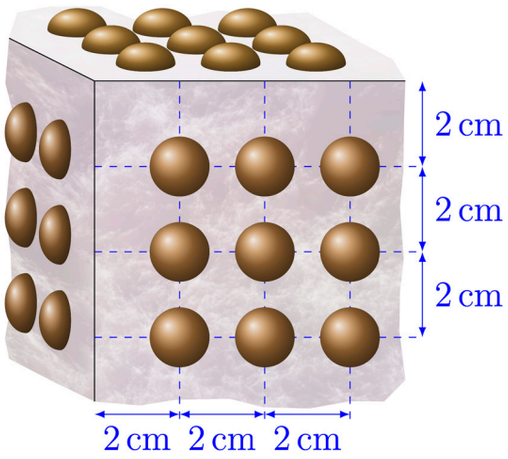
27. 妈妈为凯伦制作了一个生日蛋糕。蛋糕的顶面和 4 个垂直侧面铺好糖霜后，形成了一个边长为 20 cm 的立方体。
小达要用巧克力豆装饰这个蛋糕。
他在糖霜上将巧克力豆排列成方形网格的图案，巧克力豆中心之间的距离为 2 cm，靠近立方体边缘的巧克力豆的中心距离边缘也是 2 cm。
如图所示是这个蛋糕的一个角。
请问小达一共用了多少颗巧克力豆来装饰凯伦的蛋糕？



Karen's mother made a cake for her birthday. After it was iced on the top and the 4 vertical faces, it was a cube with 20 cm sides. Darren was asked to decorate the cake with chocolate drops.
He arranged them all over the icing in a square grid pattern, spaced with centres 2 cm apart. Those near the edges of the cube had centres 2 cm from the edge.
The diagram shows one corner of the cake.
How many chocolate drops did Darren use to decorate Karen's cake?

---

28. 我从 1、3、5、7、9、$\cdots$、105 中选出三个互不相同的数，然后将它们相加。
请问我一共可以得到多少个不同的和？

I choose three different numbers out of this list and add them together:

$$1, 3, 5, 7, 9, \ldots, 105$$

How many different totals can I get?

---

29. A 和 W 两个运动俱乐部决定合作组建一支队伍参加区域锦标赛。这支队共有 11 位短跑选手，其中有 5 位来自 A 俱乐部， 6 位来自 W 俱乐部。
在 4×100 接力赛中，他们同意由 A 俱乐部的两位短跑选手和 W 俱乐部的两位短跑选手组成一支接力队参加。
请问一共有多少种可能的接力队组合方式？

The athletics clubs of Albury and Wodonga agree to send a combined team to the regional championships. They have 11 sprinters on the combined team, 5 from Albury and 6 from Wodonga.
For the 4×100 metre relay, they agree to have a relay team with two sprinters from the Albury club and two sprinters from the Wodonga club.
How many relay teams are possible?

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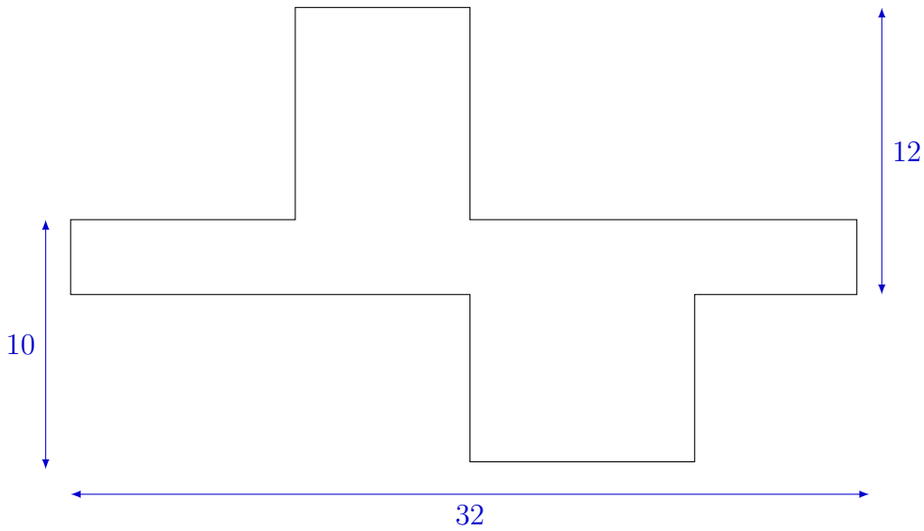
30. 下图是一个长方体的展开图，已经标记出它的一些尺寸，以 cm 为单位。



请问这个长方体的体积是多少 cm^3？

The diagram shows a net of a rectangular prism with some dimensions, in centimetres, given.

What is the volume of the rectangular prism in cubic centimetres?